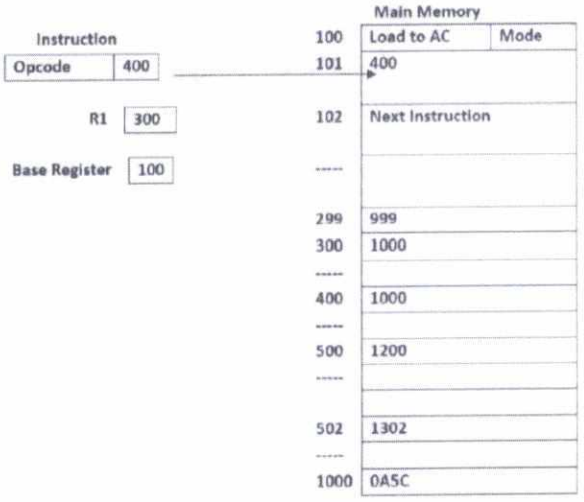
Roll Number: .....
Name: .....

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science &Engineering

B. E.(COE/CSE) : Auxiliary	Course Code: UCS510 /UCS319
	Course Name: Computer Architecture and Organization
Date: 28/8/2026	Time: 5:30 PM – 8:30 PM
Time: 3 Hours, M. Marks: 100	Faculty: SHI

Note: All questions carry equal marks. Attempt any 5 questions in order.
Assume the missing data.

<table>
<tr><td>Q1(A)</td><td>Design an ALU circuit with selection variable S_0 and two 2-bit data inputs A and B. The circuit generates the following four arithmetic micro-operations in conjunction with the input carry C_{in}.
<table>
<tr><th>S0</th><th>Cin</th><th>Micro-operation</th></tr>
<tr><td>0</td><td>0</td><td>A+B</td></tr>
<tr><td>0</td><td>1</td><td>A-B</td></tr>
<tr><td>1</td><td>0</td><td>Transfer A</td></tr>
<tr><td>1</td><td>1</td><td>Increment A</td></tr>
</table></td><td>10</td></tr>
<tr><td>Q1(B)</td><td>Convert the following numbers into IEEE-754 single-precision (32-bit) floating point representation.
i. $(C3.125)_{16}$ ii. $(75.25)_{10}$</td><td>10</td></tr>
<tr><td>Q2(A)</td><td>Assume 5-bit registers are used to represent signed numbers. Using Booth's multiplication algorithm, demonstrate the complete step-by-step multiplication process for a multiplicand of 5_{10} and a multiplier of 9_{10}.</td><td>10</td></tr>
<tr><td>Q2(B)</td><td>What are the different addressing modes? Explain each addressing mode with a suitable example. Also, discuss the advantages of using different addressing modes.</td><td>10</td></tr>
<tr><td>Q3(A)</td><td>Consider a non-pipelined processor operating at a clock frequency of 2.6 GHz with an average CPI of 5. The processor is upgraded to an 8-stage pipelined architecture, but the clock frequency is reduced to 2 GHz due to internal pipeline overhead. If a stall delay of 1 ns occurs after each pipeline stage, calculate the speedup achieved by the pipelined processor over the non-pipelined processor. Show all calculation steps clearly.</td><td>10</td></tr>
<tr><td>Q3(B)</td><td>What are the different types of pipeline hazards in a processor? Explain each type with a suitable example and describe the techniques used to resolve or minimize each hazard.</td><td>10</td></tr>
<tr><td>Q4(A)</td><td>A computer system has a main memory size of 2 MB, a cache size of 32 KB, and a block size of 128 bytes. The cache uses a 4-way set-associative mapping. Find:
1. The total number of cache blocks.
2. The total number of sets in the cache.
3. The number of bits required for:
o Block offset
o Set index
o Tag</td><td>10</td></tr>
</table>

Q4(B)	A computer system has a two-level cache hierarchy. The L1 cache has a hit time of 2 ns and a hit ratio of 92%, while the L2 cache has a hit time of 10 ns and a hit ratio of 85%. The main memory access time is 70 ns. Assume that the L2 cache is accessed only when the L1 cache misses, and the main memory is accessed only when the L2 cache misses. Calculate the average memory access time (AMAT). Show all calculation steps clearly.	10
Q5(A)	Consider a 16-bit processor in which the main memory appears as:  Determine the effective address and operand to be loaded for the Indirect, PC relative, Base, Register Indirect, and Auto decrement using R1 addressing modes.	10
Q5(B)	a) Differentiate between memory references instructions and register reference instructions with one example of both. (4 differences only) b) Differentiate between RISC and CISC. (4 differences only) c) Differentiate between BUN and BSA instruction (2 differences only)	10
Q6(A)	An I/O device needs to transfer a 12 MB file directly to main memory using a DMA controller. Explain the step-by-step working of DMA, including the functions of all relevant registers and the control signals involved. Also explain the need and advantages of DMA over programmed I/O. Support your answer with a neat and clearly labeled diagram.	10
Q6(B)	A DMA controller is used to transfer a block of 4096 bytes from an I/O device to main memory. If each DMA transfer cycle takes 400 ns, calculate: 1. The total time required to complete the DMA transfer. 2. The data transfer rate in MB/s. Show all calculation steps clearly.	10